

PORTO VELHO, Brazil

WMO: 828240

Lat: 8.709S

Lon: 63.902W

Elev: 90

StdP: 100.25

Time Zone: -4.00 (W04)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 18.9	(c) 20.1	(d) 12.2	(e) 8.9	(f) 28.2	(g) 14.7	(h) 10.6	(i) 28.0	(j) 6.6	(k) 27.7	(l) 5.5	(m) 27.7	(n) 2.0	(o) 180	(p) 0.287

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	10.8	36.0	24.0	35.1	24.3	34.2	24.6	28.1	31.0	27.6	30.8	27.2	30.5	3.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.2	29.2	27.0	22.9	29.0	26.2	21.9	28.4	90.7	31.4	88.5	31.0	86.7	30.3	33.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 6.0	(b) 5.0	(c) 4.3	(e) 16.3	(f) 37.3	(g) 1.9	(h) 1.0	(i) 14.9	(j) 38.0	(k) 13.8	(l) 38.6	(m) 12.7	(n) 39.1	(o) 11.3	(p) 39.8		
			DB	WB	14.2	29.2	2.6	1.6	12.3	30.4	10.8	31.3	9.3	32.2	7.4	33.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	27.0	27.0	27.3	27.5	27.1	27.1	27.2	28.2	28.4	28.1	27.7	27.2
	DBStd	1.51	1.18	1.20	1.17	1.35	1.57	1.70	1.86	1.53	1.58	1.30	1.31	1.26
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	0	0	0	0	0	1	0	0	0	0	0	0
	CDD10.0	6382	527	476	536	526	530	513	532	563	552	562	531	533
	CDD18.3	3341	269	242	278	276	272	264	274	304	302	304	281	275
	CDH23.3	29652	2054	1867	2231	2362	2331	2426	2712	3186	3042	2824	2452	2164
(14) Wind	WSAvg	1.4	1.5	1.4	1.4	1.3	1.4	1.3	1.4	1.4	1.5	1.5	1.4	1.4
	PrecAvg	1961	325	272	285	198	111	29	34	36	93	170	195	285
	PrecMax	3160	601	433	521	358	232	138	142	99	205	310	316	592
	PrecMin	780	67	67	65	48	20	5	0	0	1	14	68	42
	PrecStd	635	123	100	109	87	56	29	41	31	59	62	66	137
	0.4%	DB	33.8	33.4	33.9	34.1	34.1	35.0	36.1	37.2	36.9	36.1	35.0	34.1
	0.4%	MCWB	26.2	26.2	26.5	26.5	25.6	24.6	23.0	22.6	24.8	25.4	26.0	26.4
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	32.1	32.1	32.8	33.1	33.0	33.8	35.0	36.0	35.9	34.9	33.9	32.9
	2%	MCWB	26.0	25.9	26.1	26.5	25.9	25.0	23.0	23.0	24.8	25.5	26.1	26.2
	5%	DB	31.1	31.2	31.8	32.1	32.1	33.0	34.0	35.0	34.9	33.8	32.9	31.8
	5%	MCWB	26.0	25.9	26.0	26.4	25.9	24.7	23.2	23.3	24.9	25.6	25.9	26.0
	10%	DB	30.1	30.1	30.5	31.0	31.0	32.0	33.0	34.0	33.8	32.5	31.6	30.6
	10%	MCWB	25.7	25.7	25.9	26.3	25.9	24.9	23.6	23.6	24.9	25.5	25.9	25.9
	10%	MCDB	28.7	28.6	29.0	29.6	29.4	29.8	29.6	30.1	30.5	30.2	29.6	29.1
(35) Mean Daily Temperature Range	MDBR	7.2	7.1	7.4	7.7	8.0	9.5	11.4	12.1	10.8	9.5	8.4	7.5	
	5% DB	MCDBR	8.5	8.4	8.8	8.9	9.3	10.9	12.8	13.5	12.4	11.0	9.9	9.1
	5% DB	MCWBR	3.4	3.2	3.4	3.6	3.7	4.0	3.8	3.6	3.5	3.5	3.6	3.5
	5% WB	MCDBR	7.7	7.5	7.8	8.1	8.1	9.4	10.6	11.2	10.6	9.6	8.7	8.2
	5% WB	MCWBR	3.6	3.4	3.7	4.0	4.1	4.6	4.6	3.8	3.7	3.8	3.8	3.7
	taub	0.398	0.389	0.393	0.396	0.397	0.401	0.416	0.599	0.672	0.520	0.452	0.422	
	taud	2.507	2.541	2.538	2.497	2.483	2.471	2.402	1.861	1.712	2.124	2.324	2.425	
(42) Clear-Sky Solar Irradiance	Ebn at Noon	943	946	928	896	866	847	840	713	689	820	885	917	
	Edh at Noon	114	110	108	108	104	103	112	201	244	165	136	123	
	RadAvg	4.47	4.38	4.51	4.59	4.46	4.83	5.24	5.15	4.96	5.02	4.68	4.38	
(45) All-Sky Solar Radiation	RadStd	0.23	0.26	0.21	0.23	0.27	0.31	0.35	0.40	0.43	0.19	0.24	0.29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-2.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page